



Computational fluid mechanics researcher building physics-informed ML surrogates for PDE systems, with large-scale field-data pipelines on HPC.

Education

University of Pittsburgh — PhD, Civil and Environmental Engineering	Expected 2031
University of Southern California — MS, Aerospace & Mechanical Engineering (AI and Machine Learning)	2026
Shanghai Jiao Tong University — BSE, Mechanical Engineering	2025

Technical skills

Machine learning	PyTorch, TensorFlow, physics-informed neural nets, Transformers, PDE surrogate modeling
Programming	Python (NumPy, SciPy, PyTorch), MATLAB, C/C++
Simulation	ANSYS, Abaqus/CAE, COMSOL, SolidWorks; multiphase CFD, finite element analysis
Computing	HPC (Pitt, SJTU), HDF5, Zarr, Git, pytest, Streamlit, Linux/Bash, PowerShell

Simulation and machine learning

Data-Driven Turbulence Research 2026–present

Graduate Student Researcher, University of Pittsburgh

- Built a resumable pipeline assembling full 1024^3 JHTDB velocity fields from 96 MiB chunked requests (100 MiB cap), with checksum tiles, retry/backoff, and Blosc Zarr output at 9.3 GB/frame.
- Ran slab-streamed FFT filtering and spectral derivatives at five scales on 16 workers; 92 GiB peak, 23 GiB/scale.
- Derived subfilter transfer fields ($\Pi = \tau_{ij}\partial_j \bar{u}_i$, $\bar{S}$, resolved and full work) over a six-regime partition, quantifying forward vs. backscatter per regime.
- Added QA gates on divergence, energy identity, and regime closure, plus a read-only Streamlit viewer.

Physics-Informed Transformer for 2D Navier–Stokes 2025

Team Course Project (4 members), AME-505, University of Southern California

- Built the PDEBench 2D Navier–Stokes data pipeline: HDF5 extraction of 512×512 velocity and forcing fields, 200-frame truncation, and a dataloader supplying 204,800 samples from 1,024 sensors.
- Cut ~50M space–time points to a stride-16 grid plus 4,096 random collocation points to fit one RTX 3090.
- Implemented an encoder-only Transformer (4 layers, 4 heads, 128-dim) with Fourier positional encoding and a weighted data/momentum/continuity/BC loss, reaching 0.03–0.06 relative L2 at ~10 ms inference.

Computational mechanics and CFD

Computational Cell Biomechanics 2024–present

Independent Researcher, Shanghai Jiao Tong University

- Built 2D Neo-Hookean FE models of AFM indentation in Abaqus/CAE with explicit actin, microtubules, and $A\beta 42$ pores.
- Reproduced the measured 4–14% modulus drop from simulated force–displacement curves; under review at PNAS.

Computational Multiphase Flow 2022–2023

Research Assistant, Shanghai Jiao Tong University

- Ran gas–vapor–water multi-fluid CFD of ventilated supercavitating flows at 20 and 140 m/s ($Fr = 15.1$ and 105.3), spanning gravity-dominated and gravity-negligible closure, with per-phase SST $k-\omega$.
- Established radial distribution laws of velocity, volume fraction, and superficial velocity for all three phases.
- Derived a shear-layer gas loss model from the shear flow and phase distributions, validated against water-whirling-arm and water-tunnel experiments; published in *Physics of Fluids*.

Publications and honors

- Zou, W., Liu, T., **Gao, X.**, Shi, Y. “Multiphase Flow Characteristics and Gas Loss in the Shear Layer on a Ventilated Supercavity Wall.” *Physics of Fluids* 35, 043318 (2023). DOI: 10.1063/5.0141678
- Jin, H., Suo, Q., Ma, S., **Gao, X.**, et al. “Beta-amyloid42 Oligomers Soften Neuronal Cells by Reorganization of Cytoskeleton.” Under review, *PNAS* (2026).
- Four conference papers on ventilated supercavitation, second author on two: ACFM 2023; CCTAM, NCFM, ICHD 2022.
- Outstanding Graduate of Shanghai (top 3% of SJTU) 2025; Wu and Taihu research scholarships 2023; Excellent TA 2024.